

Autonomous Hovercraft Project

COEN/ELEC 290 – Engineering Team Design Project
Concordia University

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Table of Contents

<i>Autonomous Hovercraft Project.....</i>	<i>1</i>
<i>Project Overview</i>	<i>2</i>
<i>Project Requirements and Objectives.....</i>	<i>2</i>
<i>Main Components and Technologies</i>	<i>2</i>
<i>System Operation</i>	<i>3</i>
<i>Hardware Integration</i>	<i>3</i>
<i>Development and Testing.....</i>	<i>4</i>
<i>My Contribution.....</i>	<i>4</i>
<i>Final Result.....</i>	<i>5</i>
<i>Skills and Experience Gained</i>	<i>5</i>

Project Overview

The Autonomous Hovercraft was a course project completed by a team of six students for COEN/ELEC 290 at Concordia University. The goal of the project was to design, build, and test a functional hovercraft capable of navigating autonomously through a designated maze while detecting walls and obstacles.

The project required us to combine mechanical construction, sensors, embedded electronics, propulsion, steering, and autonomous control into one complete system. The final hovercraft was able to generate sufficient lift, move through the test environment, detect its surroundings, steer, and navigate autonomously.

Project Requirements and Objectives

The main requirements of the project were to:

- Design and construct a functional hovercraft.
- Generate enough lift to reduce friction with the ground.
- Provide forward propulsion and directional control.
- Detect nearby walls and obstacles using sensors.
- Use an embedded controller to process sensor information.
- Navigate autonomously through the required test environment.
- Integrate the mechanical, electrical, sensing, and control systems.
- Test and improve the design until a functional final prototype was achieved.

Main Components and Technologies

The hovercraft used the following main components and technologies:

- Arduino-based microcontroller
- Ultrasonic distance sensors
- Infrared sensing
- MPU6050 inertial measurement unit
- Lift fan
- Thrust/propulsion fan
- Servo motor for steering
- Battery and onboard power system
- Motor-control electronics
- Custom hovercraft chassis

- Flexible plastic hovercraft skirt
- Embedded C/C++ firmware

System Operation

The hovercraft used a lift fan to force air underneath the chassis and into a flexible plastic skirt. This created an air cushion between the hovercraft and the ground, reducing friction and allowing the vehicle to move more easily.

A separate thrust fan provided forward propulsion. A servo-controlled steering mechanism redirected the thrust to turn the hovercraft left or right.

Ultrasonic and infrared sensors were used to detect walls and obstacles around the vehicle. The sensor information was processed by the Arduino-based controller, which used the navigation logic to determine the required steering action.

The embedded control system also incorporated an MPU6050 inertial measurement unit to provide motion and orientation information. During autonomous navigation, the system could maintain its position between walls, detect openings, initiate turns, and return to normal forward navigation after completing a turn.

Hardware Integration

The mechanical and electrical components had to be integrated carefully because the hovercraft was sensitive to weight and weight distribution.

The lift system, thrust fan, steering servo, sensors, controller, power system, and other components were mounted onto the chassis while keeping the vehicle as balanced as possible. The flexible skirt was installed underneath the chassis to contain the airflow produced by the lift fan.

Component placement was adjusted during development to improve the hovercraft's balance, stability, and overall performance.

Development and Testing

The hovercraft was developed through repeated construction, testing, adjustment, and retesting.

The team tested different parts of the system, including:

- Lift performance
- Propulsion
- Steering response
- Sensor measurements
- Sensor calibration
- Weight distribution
- Hovercraft stability
- Obstacle detection
- Autonomous navigation

During testing, mechanical adjustments were made to improve balance and hovering performance. Sensor positions and calibration were also adjusted to improve the reliability of the navigation system.

The complete system was then tested as an integrated platform before the final demonstration.

My Contribution

My main contribution to the project was focused on the physical construction, mechanical integration, calibration, and testing of the hovercraft.

My work included:

- Fabricating and assembling the hovercraft chassis and body.
- Constructing and installing the flexible plastic skirt.
- Assisting with the integration of the lift, propulsion, steering, and sensing components.
- Adjusting component placement and weight distribution to improve stability.
- Assisting with infrared and ultrasonic sensor calibration.

- Performing practical testing of the hovercraft.
- Troubleshooting mechanical and stability problems.
- Supporting final system integration and testing.

The embedded navigation software was developed as part of the overall team project. My primary responsibilities were related to the physical platform, hardware integration, calibration, stability, and experimental testing.

Final Result

The final prototype successfully demonstrated the main requirements of the project.

The hovercraft was able to:

- Produce stable lift using the air-cushion system.
- Move forward using the thrust fan.
- Steer using the servo-controlled directional system.
- Detect surrounding walls and obstacles.
- Integrate the mechanical, electrical, sensing, and control systems.
- Navigate autonomously through the designated test environment.

The final demonstration showed that the different subsystems could work together successfully as one complete autonomous vehicle.

Skills and Experience Gained

Through this project, I gained practical experience in:

- Engineering prototyping and fabrication
- Mechanical assembly
- Hardware integration
- Sensor calibration
- Ultrasonic and infrared sensing
- Weight distribution and vehicle stability
- Embedded systems
- Hardware/software interaction
- Experimental testing
- Troubleshooting
- Iterative engineering design
- Engineering teamwork

The project also gave me experience working on a multidisciplinary system where mechanical design, electronics, sensing, and embedded control all had to work together for the final product to function successfully.

